

Clique and chromatic numbers of binary subspace orthogonality graphs

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Abstract

We determine the clique number of the orthogonality graph on the nonzero subspaces of $\mathbb{F}_2^n$ and establish that its chromatic number in dimension six is 15. The clique upper bound follows from a radical and quotient argument. The chromatic upper bound is certified by a complete coloring of all 2824 nonzero subspaces and an independent check against coordinate orthogonality. For the induced graph on lines, we prove a weighted independent-set bound and construct a 13-coloring, giving $12 \leq \chi(\Gamma_6) \leq 13$. Thus the full subspace graph requires at least two more colors than its line subgraph. We describe the geometry of the fixed-clique color lists and a dimension reduction for extending the search to dimension seven.

1 Introduction and statements of the results

Let $V_n = \mathbb{F}_2^n$ with the standard bilinear form $\langle x, y \rangle = \sum_{i=1}^n x_i y_i$. The graph $\mathcal{O}_n^*$ has the nonzero linear subspaces of V_n as its vertices. Distinct vertices U, W are adjacent when every vector of U is orthogonal to every vector of W . Its induced subgraph on one-dimensional subspaces is denoted by Γ_n . Over $\mathbb{F}_2$, each line has a unique nonzero vector.

Nikandish [1] asks for the clique number in Problem 4.2 and the chromatic number for $n \geq 6$ in Problem 4.3. Put $N(r) = \sum_{d=1}^r \begin{bmatrix} r \\ d \end{bmatrix}_2$, where $\begin{bmatrix} r \\ d \end{bmatrix}_2$ is the Gaussian binomial coefficient, and set $N(0) = 0$. Our results are

$$\omega(\mathcal{O}_n^*) = \max\{n, N(\lfloor n/2 \rfloor) + (n \bmod 2)\} \quad (n \geq 1),$$

and

$$\chi(\mathcal{O}_6^*) = 15, \quad 12 \leq \chi(\Gamma_6) \leq 13.$$

The first formula answers Problem 4.2 in all dimensions. The second answers the dimension-six case of Problem 4.3 and quantifies the separation between coloring the line graph and coloring the full subspace graph.

The clique argument is algebraic. The full-graph chromatic upper bound uses an explicit finite certificate, while the line lower bound follows from the geometry of pairwise nonorthogonal vectors. The verification details and the scope of the Lean formalization are collected in Section 7.

2 The exact clique number

Theorem 2.1. *For every positive integer n ,*

$$\omega(\mathcal{O}_n^*) = \max\{n, N(\lfloor n/2 \rfloor) + (n \bmod 2)\}.$$

The coordinate lines give the lower bound n . For $r = \lfloor n/2 \rfloor$, the subspace generated by $e_1 + e_2, \dots, e_{2r-1} + e_{2r}$ is totally isotropic. Its nonzero subspaces form a clique of size $N(r)$; when n is odd, the line generated by e_n can be added. These are the lower-bound constructions of [1].

For the upper bound, let $\mathcal{C}$ be a clique and let R be the sum of the radicals of its members. Then $R \subseteq R^\perp$. Writing $r = \dim R$, the images of the members not contained in R form mutually orthogonal, nonzero nondegenerate subspaces of $R^\perp/R$. They are linearly independent as subspaces, so there are at most $n - 2r$ of them. The members contained in R number at most $N(r)$. Thus

$$|\mathcal{C}| \leq N(r) + n - 2r, \quad 0 \leq r \leq \lfloor n/2 \rfloor.$$

The successive increments of $N(r)$ are nondecreasing, so the maximum of this expression occurs at an endpoint and gives Theorem 2.1. The complete radical/quotient proof is retained in the existing formal module cited in Section 7; this section is the integration starting point for the expanded algebraic presentation.

3 The line graph in dimension six

Lemma 3.1. *An independent set in Γ_6 containing t even-weight vectors and o odd-weight vectors satisfies $3t + 4o \leq 19$. Its cardinality is at most six; cardinality six requires $(t, o) = (5, 1)$.*

Proof. Write $j = (1, 1, 1, 1, 1, 1)$. The even-weight vectors lie in $j^\perp$, whose restricted form has radical $\text{span}(j)$ and rank four. If $o = 0$, the Gram matrix of the selected vectors is $J_t + I_t$, of rank t for even t and $t - 1$ for odd t . Thus $t \leq 5$.

If $o > 0$, fix an odd selected vector a . For each even selected vector e , the vector $z_e = e + a$ belongs to $a^\perp$. These vectors are orthonormal and span a nondegenerate t -space A . The ambient space $a^\perp$ is nondegenerate of dimension five. The span D of the vectors $u + a$, over odd selected vectors u , is totally isotropic and orthogonal to A inside $a^\perp$. Hence $2 \dim D \leq 5 - t$. The odd selected vectors are distinct members of $a + D$, so $o \leq 2^{\dim D}$. Consequently $o \leq 4$ for $t = 0, 1$, $o \leq 2$ for $t = 2, 3$, and $o \leq 1$ for $t = 4, 5$. These bounds prove both claims. $\square$

Theorem 3.2. $12 \leq \chi(\Gamma_6) \leq 13$.

Proof. There are 31 nonzero even-weight and 32 odd-weight vectors. Their total weight, using weights three and four respectively, is 221. By Lemma 3.1, every color class has weight at most 19, giving $\chi(\Gamma_6) \geq \lceil 221/19 \rceil = 12$. For the upper bound, interpret an integer x as the vector whose coordinate i is its i th binary digit. The certificate `results/line-coloring.json` partitions the 63 nonzero vectors into 13 classes. The independent checker verifies that every pair of distinct vectors in each class has dot product one. $\square$

Any hypothetical 12-coloring has a class of size six. By the lemma it consists of an odd vector a and five even vectors e . The vectors a and $e + a$ form an orthonormal basis. An isometry therefore maps this class to $\{1, 3, 5, 9, 17, 33\}$. Fixing this class reduces the remaining question to an 11-coloring of the induced graph on the other 57 vectors.

4 The full chromatic number in dimension six

Theorem 4.1. $\chi(\mathcal{O}_6^*) = 15$.

Proof. Let $T = \text{span}(3, 12, 48)$ in binary integer coordinates. The form vanishes on T , so its seven lines, seven planes and T itself form a 15-clique. For the upper bound, the file `results/full-15-coloring.json` gives a basis and a color in $\{0, \dots, 14\}$ for every nonzero subspace. The independent checker reconstructs each span and verifies basis independence, distinctness, and the dimension counts

$$63, \quad 651, \quad 1395, \quad 651, \quad 63, \quad 1.$$

These equal $\begin{bmatrix} 6 \\ d \end{bmatrix}_2$ for $1 \leq d \leq 6$, proving completeness of the catalogue. For all 3 986 076 pairs of distinct entries it tests orthogonality on their basis vectors. Bilinearity identifies this test with graph adjacency. All 44 968 orthogonality edges have distinct endpoint colors. This establishes the upper bound and therefore equality. $\square$

To find the certificate, fix distinct colors on the 15 nonzero subspaces W of T . The forbidden colors for a vertex $U \not\subseteq T$ are exactly those $W \subseteq H(U) = U^\perp \cap T$. Since $T = T^\perp$, the possible dimensions of $H(U)$ are 0, 1, 2, and the resulting list sizes are 15, 14, 11.

A vertex whose residual degree is smaller than its list size can be deleted and restored greedily after the remaining graph is colored. Repeated deletion removes 729 uncolored vertices and leaves 2 080 vertices with 39 400 edges. The list-coloring encoding has 28 600 variables and 627 510 clauses. A satisfying assignment was extended in reverse deletion order to produce the full certificate checked above.

5 Geometry of the precoloring

The color labels are the nonzero subspaces W of the fixed totally isotropic space T . The relation $W \subseteq H(U)$ determines exactly which fixed clique colors are excluded at U . The additional invariant $U \cap T$ records how U meets the isotropic space itself.

The reproducibility package groups the archived coloring by $(\dim U, H(U), U \cap T)$. Outside the clique, 240 such profiles occur, of which 139 receive more than one color in this certificate. These data identify where a structural construction would need to refine or reorganize the present choices. They do not exclude the existence of a different coloring determined by a smaller collection of geometric invariants. The files `results/coloring-geometry.json` and `develop/analyze_geometry.py` provide the complete counts and color labels.

6 Higher dimensions and open problems

Proposition 6.1. *Let K be a fixed k -clique in $\mathcal{O}_n^*$. If $N(n - d) < k$, every uncolored d -dimensional vertex can be deleted by the degree-versus-list rule. Consequently, k -colorability of the residual instance after such deletions is equivalent to k -colorability of the full graph.*

Proof. For a d -space U , the orthogonal complement has dimension $n - d$. Every neighbor of U is a nonzero subspace of $U^\perp$, so $\deg U \leq N(n - d)$. If f fixed clique vertices are adjacent to U , then its residual degree is at most $N(n - d) - f$, while its list size is $k - f$. The stated inequality therefore guarantees an available color on restoring U . Apply this argument in reverse deletion order. $\square$

For $n = 7$, the nonzero subspaces of $T = \text{span}(3, 12, 48)$ together with $\text{span}(64)$ give a 16-clique. Since $N(3) = 15 < 16$, Proposition 6.1 removes all 14 606 vertices of dimension at least four from the 29 211-vertex graph. The remaining uncolored instance has 14 589 vertices and 953 057 edges. The direct pairwise list-color encoding would have 214 355 variables and 12 984 679 clauses. These exact instance sizes guide further geometric reduction and resource-bounded search.

Problem 4.3 remains open for $n \geq 7$, as does Problem 4.4 of [1]. Within dimension six, the exact value of $\chi(\Gamma_6)$ and a structural description of an optimal full-subspace coloring are the immediate questions.

7 Computational certificates and formal verification

The positive coloring certificates are checked using integer arithmetic and the coordinate definition of orthogonality. The full checker imports neither the search program nor a SAT solver. It establishes enumeration completeness from distinctness and Gaussian counts, then checks every pair of vertices. Three deliberately invalid certificates, respectively omitting a vertex, duplicating a subspace, and identifying the colors of adjacent coordinate lines, are rejected. The line-coloring checker independently verifies its partition and all within-class pairs.

The radical/quotient clique theorem has an existing Lean formalization. Its source is `formal/NikandishClique.lean`, with theorem `NikandishClique.result` in its documented namespace. The exact upstream revision, Lean version and dependencies are recorded in `formal/UPSTREAM.json`. The new chromatic certificate is connected to the kernel-checked theorem `chromatic_number_six` in `formal/Chromatic/Verify.lean`. The Mac Studio build, source hashes and axiom scope are recorded in `results/lean-verification.json`; the independent integer checker remains a separate verification of the JSON witness.

The public package [2] contains the literal witnesses, independent checkers, search records, and SHA-256 bindings. The new geometric profiler checks every assigned color against its $H(U)$ -determined list. The dimension-seven reduction is also replayed in dimension six, where it reproduces the established core size and Boolean encoding counts.

References

- [1] R. Nikandish, *Annihilating-Ideal Graphs and Orthogonality Graphs over $\mathbb{F}_2$* , arXiv:2609.22769v1, 2026.
- [2] H. Ma, R. Nikandish and W. Zhang, *Binary subspace orthogonality graphs: research and reproducibility workspace*, <https://github.com/the-omega-institute/binary-subspace-orthogonality>.